

Let's take a walk through a GraceDB page

<https://gracedb.ligo.org/superevents/S251108fi/view/>

<https://gracedb.ligo.org/superevents/S251112ad/view/>

Handling sky localization maps

	Flat Maps	Multi-Order Maps (HEALPix Multi-Order, MOC)
Underlying Pixelization	HEALPix with fixed resolution (one Nside)	HEALPix with variable resolution (multi-order)
Storage Size	Always full sky → large files even if probability is localized	Only stores pixels that contain probability → 10–100× smaller
Resolution	Uniform resolution everywhere (wasteful in low-probability regions)	High resolution only where needed (e.g., 0.1° near source, degrees elsewhere)
Probability Representation	Fixed grid; downsampling loses information	Exact lossless representation at arbitrary resolution
Typical File Size	50–200 MB per event	50–500 kB per event